

A Signal-Based Forensic Framework for Explainable AI-Generated Video Detection

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Abstract

The rapid advancement of AI-based video generation models has made it increasingly difficult to distinguish synthetic videos from real-world footage. While many existing detection approaches rely on deep learning classifiers, these methods often lack transparency and fail to generalize across unseen models.

This paper presents a signal-based forensic framework for estimating the likelihood that a video is AI-generated using explainable spatial and temporal cues. The proposed system analyses noise variance, temporal noise correlation, and high-frequency energy extracted from video frames to produce an interpretable risk score.

Unlike black-box neural detectors, the framework emphasizes transparency, reproducibility, and model-agnostic behaviour. Experimental observations show that the system correctly assigns low AI risk to CGI and animated content while identifying anomalous patterns commonly present in AI-generated videos.

This work highlights the potential of classical signal processing techniques as a complementary approach to modern AI video forensics.

1. Introduction

The proliferation of AI-generated visual content has introduced significant challenges in media authenticity, digital forensics, and content moderation. Recent video generation models are capable of producing highly realistic sequences, often indistinguishable from real footage to human observers. This raises concerns related to misinformation, deepfake misuse, and trust in digital media.

Most existing AI-video detection systems rely on supervised deep learning models trained on specific datasets. While effective in controlled settings, such approaches often suffer from poor generalization, lack of interpretability, and vulnerability to model drift as generation techniques evolve.

This project explores an alternative direction: a signal-based forensic approach that analyses low-level spatial and temporal characteristics inherent to video data. By focusing on explainable noise patterns and frequency-domain behaviour, the system aims to provide transparent indicators of AI generation without relying on opaque neural classifiers.

The primary goal of this work is not to provide absolute classification but to estimate probabilistic risk using interpretable forensic signals.

The contributions of this work are threefold. First, it proposes a lightweight, signal-based forensic framework for AI-generated video detection that does not require model training. Second, it demonstrates how classical image and signal processing techniques can reveal artifacts introduced by neural video synthesis. Finally, it presents an explainable scoring mechanism that provides insight into why a video may be classified as AI-generated rather than relying on opaque predictions.

An open-source implementation and a publicly deployable system are provided to encourage reproducibility and further research

2. Related Work

Early approaches to AI-generated media detection primarily relied on deep learning classifiers trained on large datasets of real and synthetic videos. These methods achieved strong performance in controlled environments but often failed to generalize to unseen generation models or compression settings.

Several studies have explored spatial artifacts such as inconsistent textures, abnormal lighting, and frequency-domain irregularities introduced by generative models. Other works focused on temporal inconsistencies, including flickering, motion instability, and noise reuse across frames.

More recent research has highlighted the limitations of purely data-driven detectors, emphasizing the need for explainable and model-agnostic forensic techniques. Classical signal processing methods, including noise analysis and frequency-domain inspection, have been revisited as complementary tools for detecting synthetic media.

This project builds upon these ideas by combining spatial noise statistics and temporal correlation analysis into a unified, interpretable risk scoring framework.

3. Methodology

The proposed system follows a signal-based forensic pipeline designed to analyse low-level spatial and temporal characteristics of video frames. Instead of training a classifier, the framework extracts interpretable signals from decoded video frames and aggregates them into a probabilistic risk score.

The input video is first decoded into individual frames at a configurable frame rate. Each frame undergoes grayscale conversion and spatial filtering to isolate noise residuals. These residuals are then analysed using statistical and frequency-domain metrics.

Three primary forensic signals are extracted: noise variance, temporal noise correlation, and high-frequency energy. These signals are computed independently and later combined to estimate the likelihood of AI generation.

3.1 Noise Residual Extraction

Noise residuals are obtained by subtracting a smoothed version of each frame from the original grayscale image. Gaussian blurring is used to remove semantic structure, leaving behind high-frequency noise patterns. These residuals serve as the basis for subsequent statistical and frequency-domain analysis.

3.2 Forensic Signal Computation

Noise variance measures the overall energy of the residual signal. AI-generated videos often exhibit lower or more uniform variance due to over-smoothing during generation.

Temporal noise correlation evaluates the similarity between noise residuals of consecutive frames. Generative models may reuse noise patterns across frames, leading to higher correlation than typically observed in real camera footage.

High-frequency energy is computed using the Fourier transform of the noise residual. Real videos captured by physical sensors contain natural high-frequency grain, while synthetic videos often display abnormal frequency distributions.

4. Experimental Setup and Observations

The proposed framework was evaluated through a series of controlled experiments using publicly available video content and synthetic examples. The objective of these experiments was not to benchmark classification accuracy against large datasets, but to qualitatively observe how the extracted forensic signals behave across different categories of video content.

4.1 Test Data Selection

Three broad categories of video content were analysed:

- **Real-world camera footage**, captured using physical imaging sensors
- **AI-generated videos**, produced using modern generative video models
- **CGI and animated content**, rendered using deterministic graphics pipelines

Publicly available sample videos were used to ensure reproducibility. Notably, *Big Buck Bunny*, a well-known Blender-rendered animated film, was included to examine how the system behaves on CGI content that is not AI-generated.

Social media platforms such as YouTube, Instagram, and TikTok were intentionally excluded from testing, as these platforms do not provide access to raw video files and apply heavy compression and transcoding, which can distort forensic signals.

4.2 Experimental Configuration

Videos were decoded into individual frames using FFmpeg at configurable frame rates. Unless otherwise stated, experiments were conducted at frame rates between **10 and 30 frames per second**, balancing computational cost and signal stability.

Each frame was converted to grayscale and processed to extract noise residuals, from which the following signals were computed:

- Noise variance
- Temporal noise correlation between consecutive frames
- High-frequency energy in the frequency domain

The extracted signals were aggregated across frames and combined into a probabilistic AI-generation risk score.

4.3 Observations

The experiments revealed several consistent patterns:

- **Real-world camera footage** typically exhibited higher high-frequency energy due to natural sensor noise and lower temporal correlation between frames.
- **AI-generated videos** frequently showed smoother noise patterns, reduced variance, and elevated temporal correlation, suggesting noise reuse or over-smoothing during generation.
- **CGI and animated content**, including *Big Buck Bunny*, consistently produced **low AI-generation risk scores**.

This behaviour is expected, as CGI content is generated through deterministic rendering pipelines rather than neural synthesis. Although visually artificial, CGI does not exhibit the same noise artifacts commonly introduced by generative AI models.

These observations reinforce the importance of distinguishing between *synthetic appearance* and *AI-based generation*, a distinction that black-box detectors often fail to make.

4.4 Discussion

Rather than providing binary classifications, the framework offers an interpretable estimate based on measurable forensic signals. This approach allows users to understand *why* a video is flagged as suspicious and avoids overconfidence in ambiguous cases.

The results demonstrate that classical signal-processing techniques remain relevant and valuable when applied thoughtfully to modern AI-forensics problems.

5. Limitations and Future Work

While the proposed signal-based forensic framework demonstrates promising behaviour across multiple video categories, several limitations must be acknowledged.

5.1 Limitations

First, the system does not claim to detect all forms of AI-generated video content. Generative models are rapidly evolving, and future architectures may reduce or eliminate some of the forensic artifacts analysed in this work.

Second, the reliability of extracted signals can be affected by **heavy compression**, aggressive transcoding, or post-processing operations. Videos sourced from social media platforms often undergo multiple compression stages, which may distort or mask low-level noise patterns.

Third, the framework does not provide binary classification decisions. Instead, it outputs a probabilistic risk score. While this improves transparency and reduces false certainty, it may require additional interpretation when used in automated decision-making systems.

Finally, the current implementation focuses on a limited set of forensic signals. Although these signals are interpretable and computationally efficient, they may not capture all subtle characteristics present in complex AI-generated videos.

5.2 Future Work

Future research can extend this framework in several directions.

Additional forensic signals, such as motion inconsistency, color channel correlations, and camera response non-uniformity (CRNU), may further improve detection reliability. Hybrid approaches that combine explainable signal-based features with lightweight machine learning models could also be explored while maintaining interpretability.

Another important direction is large-scale empirical evaluation using curated datasets containing diverse AI-generated and real-world videos. Such evaluation would enable more systematic analysis of signal distributions and robustness across generation methods.

Finally, integrating this framework into real-world forensic pipelines—such as content moderation systems, media verification tools, or academic research platforms—could provide valuable feedback on usability, performance, and practical limitations.

6. Project Availability and Reproducibility

To support transparency and reproducibility, the complete implementation of the proposed signal-based forensic framework is publicly available.

The core algorithms, including frame decoding, noise residual extraction, and forensic signal computation, are implemented in Python and released as an open-source repository.

Additionally, the system is deployed as a public **Apify Actor**, enabling reproducible execution without requiring local environment setup.

Source code repository:

https://github.com/pikachusensei/aplify_ai_video_detector

Live implementation (Apify Actor):

<https://apify.com/sensei/ai-video-detector>

7. Conclusion

This work presented a signal-based forensic framework for estimating the likelihood that a video was generated using artificial intelligence. Unlike many existing approaches that rely on opaque deep learning classifiers, the proposed system focuses on interpretable spatial and temporal signals derived from low-level video characteristics.

By analysing noise variance, temporal noise correlation, and high-frequency energy, the framework produces a probabilistic risk score that reflects potential AI generation while maintaining transparency and model-agnostic behaviour. Experimental observations demonstrate that the system correctly assigns low risk to CGI and animated content while identifying anomalous patterns commonly associated with AI-generated videos.

The results highlight the value of classical signal processing techniques as a complementary tool in modern AI video forensics. Rather than replacing learning-based methods, this approach emphasizes explainability, reproducibility, and forensic reasoning—qualities that are essential in high-stakes applications such as media verification and digital forensics.

As AI-generated content continues to advance, transparent and interpretable detection frameworks will become increasingly important. This project demonstrates that meaningful forensic insight can be achieved without relying exclusively on black-box models, and it provides a foundation for future research in explainable AI-generated video detection.

8. References

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